Multiple-Impulse Spacecraft Rendezvous Optimization

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Abstract

Writing prompt: Write this last. In one paragraph, state the rendezvous problem, the model and search method used, the main quantitative result, and the most important conclusion or limitation. Leave results unspecified until you have verified them.

1 Introduction and Aim

1.1 Spacecraft Trajectory Optimization Problem

This problem discusses how a spacecraft in a distnct orbit from another object in an orbit can rendezvous at a specific location in space at a particular time with the same velocity while minimizing the amount of fuel usage. For this specific project, we will be focusing on 2 spacecrafts in orbit around Earth where one is defined as the *chaser* spacecraft and the other is the *target* spacecraft.

1.2 Project Aim and Research Question

This leads us to the question that I aim to work around:

How can a spacecraft use a small number of engine burns to rendezvous with another spacecraft orbiting Earth?

Thus, the aim of this project that follows is to develop a simulator for orbital rendezvous missions. I will begin with a simplified model in 2 dimenions and only considering the Earth as the only relevant gravitational body. I will look to minimize the following 2 errors:

Position Error:

$$J_r = |r_c(T) - r_t(T)|^2 \tag{1}$$

Velocity Error:

$$J_v = |v_c(T) - v_t(T)|^2 \tag{2}$$

In addition to position and velocity error, I will be taking into account fuel usage with the assumption that fuel usage is directly proportional to the change in velocity in the x, y, and z directions. These three will be combined into the following cost function:

$$J = \omega_r J_r + \omega_v J_v + \omega_{\Delta v} \sum_i |\Delta v_i| \tag{3}$$

After testing with the 2 dimensional model, the aim will be to expand to 3 dimensions and will consider the Moon, Sun and Jupiter as gravitationally relevant bodies to the motion of the spacecrafts.

2 Physical Model and Problem Definition

2.1 Assumptions and equations of motion

To begin, the following assumptions will be used for the 2 dimensional model

- Planar motion in an Earth-centred inertial Cartesian frame.
- Central gravity with constant gravitational parameter μ.
- An uncontrolled target and a chaser with instantaneous burns.
- Fuel usage is directly proportional to change in velocity of a burn.

For each spacecraft, position is defined as $\mathbf{r} = \begin{bmatrix} x \\ y \end{bmatrix}$ and velocity is defined as $\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$:

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\mu \frac{\mathbf{r}}{\|\mathbf{r}\|^3}. \tag{4}$$

Equation (4) defines the ODE that the spacecraft states will be modelled after, with μ defined as the gravitational parameter and for the 2 dimensional model is the gravitational parameter of Earth:

$$\mu = GM$$
$$\mu = 3.986 \times 10^{14}$$

At burn time t_i, position is continuous and velocity changes according to

$$\mathbf{v}(t_i^+) = \mathbf{v}(t_i^-) + \Delta\mathbf{v}_i. \tag{5}$$

Equation (5) is the mathematical representation of the assumption that all burns are applied as instantaneous burns, meaning that position does not change but the velocity changes instantly right after time t_i.

For the three-dimensional model, the assumptions are extended as follows:

- Three-dimensional motion in a Sun-centred Cartesian frame, treated as inertial under the approximation that the Sun is physically fixed at the origin with zero velocity.
- Newtonian point-mass gravity from the Sun, Earth, Moon, and Jupiter, each with a constant gravitational parameter μ_b. Earth, the Moon and Jupiter move under the gravity of all four bodies. The Sun attracts the moving bodies but does not respond to their gravitational attraction. Spacecraft masses are negligible, and other gravitational bodies and non-gravitational forces are neglected.
- An uncontrolled target and a chaser with instantaneous burns. Position remains continuous across each burn, while velocity changes instantaneously.
- Total impulse magnitude, $\sum_i \|\Delta\mathbf{v}_i\|$, is used as a proxy for propellant consumption and is part of the third term for Equation 3, the cost function.

2.2 Scenario and definition of success

We can begin with defining the initial state of a spacecraft using the position and velocity vectors in 2 dimensions as follows:

$$\tilde{\mathbf{v}} = \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix} \tag{6}$$

Then applying ODE in Equation 4 to the derivative of the initial state, we obtain:

$$\tilde{\mathbf{v}'} = \begin{bmatrix} v_x \\ v_y \\ a_x \\ a_y \end{bmatrix} \tag{7}$$

$$\tilde{\mathbf{v}'} = \begin{bmatrix} v_x \\ v_y \\ -\mu \frac{x}{|r|^3} \\ -\mu \frac{y}{|r|^3} \end{bmatrix} \tag{8}$$

Then with any final time T, the number of burns and the initial state of both spacecrafts defined as shown in Equation 6 the rendezvous between the *chaser* and *target* spacecraft can be determined.

For the initial state of the 3 dimensional model with more than one significant gravitational body, we can define the acceleration using the same base ODE from Equation 4, but make the following changes to the original model:

$$\ddot{\mathbf{r}} = -\mu \frac{\mathbf{r}}{\|\mathbf{r}\|^3}.$$

$$\ddot{\mathbf{r}} = \sum_{i=0}^{n-1} -\mu_i \frac{r_i}{|r_i|^3} \tag{9}$$

Here, n is the number of gravitational bodies being considered, and where μ_i and r_i correspond to the i-th gravitational body on a 0 index. For my case, the gravitational bodies will be defined as follows:

$$[\text{Sun}, \text{Earth}, \text{Moon}, \text{Jupiter}]$$

The following gravitational parameters, μ will be used for the gravitational bodies:

Body	μ ($\text{m}^3\,\text{s}^{-2}$)
Sun	1.3271×10^{20}
Earth	3.986×10^{14}
Moon	4.9028×10^{12}
Jupiter	1.2671×10^{17}

Furthermore, we will define a successful rendezvous as follows:

- Final Separation $<= 1000m$
- Velocity Difference $<= 1ms^{-1}$

Final docking and approach are not within the scope of the project so this is sufficient for a successful rendezvous

3 Computational Method

3.1 Trajectory propagation

To begin, I use Python to calculate how the initial states of both spacecrafts change over time. The equations of motion are solved using the `solve_ivp` function from SciPy, with the `DOP853` method.

This method adjusts the size of its time steps as the trajectory is calculated. For both models, I set the relative tolerance to 10^{-6} and leave the absolute tolerance at its default value of 10^{-6}. These tolerances control the numerical integration error; they do not directly define the final separation required for a successful rendezvous.

3.1.1 Two-dimensional model

For the original 2 dimensional model, the initial states are entered in the form shown in Equation 6, with positions in metres and velocities in metres per second relative to the fixed Earth. The final time T and number of burns are also specified before the search begins. The target is first propagated from $t = 0$ to $t = T$ without any burns. I keep a continuous interpolated solution of its state using `dense_output`, allowing its position and velocity to be obtained at any time within the simulation. This target solution is reused when testing different burn sequences, since the chaser's burns do not affect the target's motion.

For the chaser, the trajectory is divided into intervals separated by the burn times. The ODE is first solved from the initial time to the first burn. At that time, the change in velocity is added to the state while position remains unchanged, as described in Equation 5. The updated state is then used as the initial state for the next interval. This process is repeated for each burn, followed by a final propagation to T. Thus, both spacecrafts are compared at the same final time even though the chaser's motion is calculated in separate intervals.

3.1.2 Three-dimensional model with multiple gravitational bodies

For the 3 dimensional model, the initial spacecraft states contain six values, $[x, y, z, v_x, v_y, v_z]$. I first obtain the initial states of Earth, the Moon and Jupiter from JPL Horizons, using the default starting epoch of 27 October 2026 at 12:00:00 UTC. These values are converted from kilometres and kilometres per second to SI units. The Sun is fixed at the origin with zero velocity throughout the simulation.

Earth, the Moon and Jupiter are then propagated under their mutual gravitational attraction and the Sun's gravity. I keep the same combined state of 24 values, with six values for each body, but set all six derivatives of the Sun's state to zero. This calculation is performed once before the rendezvous search, using the same solver and tolerances as the spacecraft propagation. A continuous interpolated solution is retained so that the location of each body can be obtained at the times requested by the spacecraft solver. I chose to hold the Sun fixed to simplify the model by neglecting its response to the other bodies.

The spacecraft states entered into the simulator describe their initial positions and velocities relative to Earth. I add Earth's initial position and velocity so that the spacecrafts and gravitational bodies can be tracked together with the Sun at the origin. At each time, I calculate the distance and direction from each body to the spacecraft and add the four gravitational contributions to obtain its acceleration:

$$\mathbf{a}(t) = -\sum_{b=1}^{4} \mu_b \frac{\mathbf{r}(t) - \mathbf{r}_b(t)}{\|\mathbf{r}(t) - \mathbf{r}_b(t)\|^3}, \tag{10}$$

where $\mathbf{r}(t)$ and $\mathbf{r}_b(t)$ are the positions of the spacecraft and gravitational body b, respectively, measured from the Sun. Their difference gives the spacecraft's displacement from that body. The target is propagated once for the search, and the chaser is propagated between burns in the same way as the 2 dimensional model. Both spacecrafts use the same precomputed body trajectories. Their masses are neglected, so neither spacecraft changes the motion of the four bodies or the other spacecraft.

3.2 Burn representation

In both implementations, a burn is stored as a pair (d_i, t_i), where d_i is a signed burn magnitude and t_i is the time of the burn. To simplify the search, burns are restricted to act along or opposite to the chaser's motion relative to Earth just before the burn. The resulting velocity change is added to the chaser's velocity, while its position remains unchanged. This is expressed as follows:

$$\Delta \mathbf{v}_i = d_i \frac{\mathbf{v}_{c/E}(t_i^-)}{\|\mathbf{v}_{c/E}(t_i^-)\|}. \tag{11}$$

A positive value of d_i applies the burn along this velocity, while a negative value applies it in the opposite direction. Thus, each burn has two search variables rather than separate velocity changes in each coordinate direction.

For the 2 dimensional model, Earth is fixed, so $\mathbf{v}_{c/E} = \mathbf{v}_c$. For the 3 dimensional model, I obtain Earth's velocity from the precomputed body solution at each burn time and subtract it from the chaser's velocity:

$$\mathbf{v}_{c/E}(t_i^-) = \mathbf{v}_c(t_i^-) - \mathbf{v}_E(t_i). \tag{12}$$

This gives the direction of the spacecraft's motion relative to Earth at the time of the burn. The same calculation is used when scoring candidate sequences and when reconstructing the selected trajectory for plotting.

3.3 Objective function and search

After each candidate burn sequence has been propagated, I calculate the squared position and velocity differences at T. The total change in velocity is calculated by adding the absolute values of the signed burn magnitudes:

$$\Delta v_{\text{total}} = \sum_{i=1}^{N} |d_i| = \sum_{i=1}^{N} \|\Delta \mathbf{v}_i\|. \tag{13}$$

These three quantities are combined using the cost function defined in Equation (3) to score each candidate burn sequence. The weights determine how much each term contributes to the search. This scaling is necessary because position error, velocity error and total change in velocity have different units and numerical sizes. For the original 2 dimensional script, the weights are fixed as

$$(\omega_r, \omega_v, \omega_{\Delta v}) = (10^{-8}, 10^{-2}, 10^{-3}), \tag{14}$$

when the states and burns are expressed in SI units. In the 3 dimensional script, the three positive weights are supplied when running the program, allowing different choices to be tested.

To search for a burn sequence, I use SciPy's `differential_evolution` function. This method starts with a population of candidate sequences, generates new candidates by combining and changing their parameter values, and keeps candidates with lower costs. For N burns, there are $2N$ parameters to search. The burn magnitude limits are 500 $\mathrm{m\,s^{-1}}$ for the original 2 dimensional model and 1000 $\mathrm{m\,s^{-1}}$ for the 3 dimensional model. Thus, for the 3 dimensional search, the bounds are

$$-1000\ \mathrm{m\,s^{-1}} \le d_i \le 1000\ \mathrm{m\,s^{-1}}, \tag{15}$$

$$60\ \mathrm{s} \le t_i \le T - 60\ \mathrm{s}. \tag{16}$$

The burn times must be strictly increasing. A candidate with repeated, out-of-order or out-of-interval burn times is assigned a cost of ∞ before propagation. The 60 second margin applies at the beginning and end of the mission; it is not a minimum spacing between successive burns. No collision-avoidance constraint is included in these scoring functions.

Both scripts use a random seed of 42 and leave the remaining optimizer settings at the SciPy defaults. In the installed SciPy version 1.17.1, these include the `best1bin` strategy, a population size of $15(2N) = 30N$, and a maximum of 1000 generations. The search can stop earlier when the spread of population costs meets the default convergence tolerance of 0.01, with zero absolute tolerance. A final local polishing step is enabled by default. The returned sequence is therefore the best sequence found by this search, rather than a guaranteed global minimum. The previously defined final separation and velocity criteria must still be checked separately; they are not enforced as constraints by the optimizer.

3.4 Trajectory reconstruction and visualization

Once a sequence has been selected, I propagate it again to produce the trajectory plots. By default, the chaser is sampled at 1000 equally spaced output times per interval, and repeated interval endpoints are removed when joining the histories. These output times are used for plotting and do not force the ODE solver to use a fixed integration step. The target's continuous solution is evaluated at the chaser's sample times to calculate their separation.

For the 2 dimensional model, the plots show the trajectories around Earth and the separation over time, with the burn locations marked. For the 3 dimensional model, the plots include separation over time and both a view of the system around the Sun and a close-up of the rendezvous around Earth. For the Earth close-up, I subtract Earth's position from each spacecraft's position at the same time. This makes the motion around Earth easier to see without changing the calculated trajectories or burn directions.

4 Simulation Checks

Writing prompt: Present checks actually performed and their outcomes. Useful checks include a circular-orbit reference case, conservation of energy and angular momentum between burns, and repeating a candidate trajectory with tighter solver tolerances. Explain what each check establishes.

- Suggested figure: reference orbit or conservation error over time.
- Report whether numerical differences are small relative to the rendezvous tolerances used in the report.

5 Experiments and Results

5.1 Baseline and best trajectory found

Writing prompt: Compare a clearly defined baseline, such as no burns or a manual strategy, with the best sequence found for the same initial conditions and final time. Fill the table with verified values and identify units.

- Plot both trajectories with burn locations and final positions.
- Plot separation and relative speed against time, marking burns.
- List each selected burn's time, signed magnitude, and direction.
- State explicitly whether both success tolerances were met.

Table 1: Rendezvous performance. Blank cells await verified results.

Strategy	e_r (m)	e_v (m/s)	Total Δv (m/s)
Baseline			
Best sequence found			

5.2 Selected comparisons

Writing prompt: Choose only comparisons that answer your research question and fit the available time. Possible variables include burn count, objective weights, or search budget. Keep other conditions fixed. For a stochastic search, use repeated seeds if assessing reliability. Report unsuccessful cases when they help explain a limitation.

6 Discussion

Writing prompt: Explain what the results mean, connecting trajectory changes to the physics rather than simply describing plots.

- Why did the chosen burn sequence improve the rendezvous attempt?
- How did timing and velocity changes affect relative motion?
- What trade-offs are supported by the experiments?
- Were remaining errors associated with numerical accuracy, search limits, burn restrictions, or objective weighting? Distinguish evidence from possible explanations.
- What do the simplified dynamics omit, and how does that limit conclusions about a real mission?

Writing prompt: Use "best solution found within the tested search" unless you have evidence of global optimality. Avoid general conclusions about burn count or fuel savings from a single case.

7 Conclusion and Future Work

Writing prompt: Answer the central question directly. Summarize the strongest quantitative finding, the main physical insight, and the principal limitation. Suggest one or two specific next steps justified by the results.

References

Writing prompt: Cite the project brief, any external theory used, numerical-method documentation, and any adapted code. Add bibliographic entries as you write.

A Reproducibility Details

Writing prompt: Record the script/version used, software dependencies, exact initial states, solver and optimizer settings, seeds, objective weights, and final burn sequence. Include supporting checks or additional plots here when they would interrupt the main argument.